

6.7 Subtracting Integers Using Number Lines

Lesson Introduction

 Benchmark	MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.		 Learning Targets	- I can subtract integers using a number line. - I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.	
 Benchmark Coherence	Building On MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.		Working Toward MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency. MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.		
 Essential Questions	- How do you subtract integers using a number line? - How are the integer chips and number-line models of integer subtraction related? - How are the patterns of adding and subtracting integers connected?				
 Lesson Narrative	In this lesson, students extend their knowledge and skills of subtracting integers using integer chips to now also using number lines. Students also discover subtraction can be rewritten as adding the additive inverse. Students make connections between the patterns of integer subtraction and addition as they solve problems throughout the lesson.				
 Component Summary	6.7.1 Warm-Up (~5 min.)	Draw integer chips to represent and evaluate expressions (SE p. 265)	6.7.3 Collaboration (20-25 min.)	Relate integer addition and subtraction (SE p. 266)	
	6.7.2 Guided Instruction (10 min.)	Subtract integers representing temperatures using a number line (SE p. 265)	6.7.4 Your Turn! (5 min.)	Practice subtracting integers using a number line (SE p. 268)	
	6.7.5 Wrap-Up (~5 min.)	Self-assess understanding of using number lines to subtract integers (SE p. 269)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Additive inverse - Zero pair		- Integer chips - Chart paper and markers		Create an anchor chart with instructions, examples, and visual models.

 Teacher Tips	Common Misconceptions Students may believe any negative value means a decreasing sum or that any negative value moves left on a horizontal number line. For example, students may believe $-9 - (-30) = -39$.	Things to Consider - Consider creating an anchor chart with instructions, examples, and visual models. Explain the chart in 6.7.2 Guided Instruction. - Consider advising students to highlight the integers being added/subtracted in 6.7.3 Collaboration.
	Literacy and Language Routines Think-Pair-Share In 6.7.3 Collaboration, have students initially complete the problems. Then have students discuss their work with a partner. The discussions will help students gain confidence and understanding.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.6.1: Reasonableness 6.7.3 Collaboration provides students multiple opportunities to assess the reasonableness of their solutions. Students discuss the problems with a partner. These discussions will help students clear up any misconceptions and deepen their understanding.
	 Differentiate Instruction	
Consider your students when planning for 6.7.3 Collaboration. Use data or informal observations from the previous lesson to inform strategic pairings for the activity to maximize engagement and flexibility. - Partner two students with different communication styles. For example, pair one student who performed well with modeling integer chips in the previous lesson with another student who effectively explains what they are seeing and excels with verbal processing. - Alternatively, students could each choose a different context before comparing their work with a partner as an expert in that context. This jigsaw approach may fit classrooms with students who work best by trying first before comparing with a partner.		
Lesson Notes:		

6.7.1 Warm-Up: Bell Work

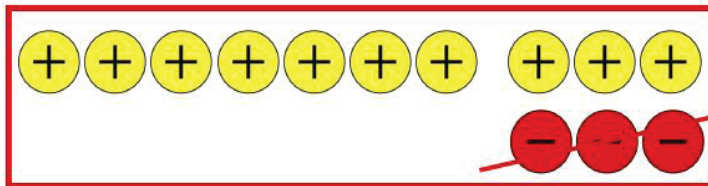
Component Overview: Students draw integer chips to represent and solve a subtraction expression. Give students the choice of either using physical integer chip manipulatives or drawing pictorial representations of integer chips.



6.7.1 Warm-Up: Bell Work

1. Draw integer chips that represent the expression below and evaluate.

$$7 - (-3)$$



$$7 - (-3) = 10.$$



Can you take away -3 from the integer chips you have represented?
How many zero pairs do you need to add to the model?

6.7.2 Guided Instruction: Subtracting Integers Using a Number Line

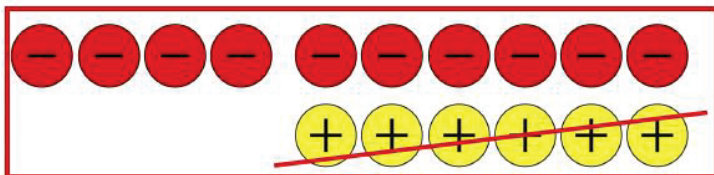
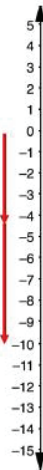
Component Overview: Students represent a temperature scenario using a vertical number line and integer chips. Students fill in guided notes that explain the process of subtracting integers using a number line.



6.7.2 Guided Instruction: Subtracting Integers Using a Number Line

1. The temperature in degrees Fahrenheit ($^{\circ}\text{F}$) of a city in Alaska was -4°F . After sunset, the temperature dropped 6°F .

- Represent the change in temperature on the number line.
- Write the subtraction number sentence that is represented from the context and on the number line.
 $-4 - 6$
- After the decrease in temperature, what was the temperature?
 $-4 - 6 = -10$
- Represent the temperature change using integer chips to verify the solution.



$$-4 - 6 = -10$$



MA.K12.MTR 2.1: Multiple Representations

Prompt students to compare the number line model to the integer chips.

- How can you show -4 degrees using a number line? How is a negative temperature represented?
- If the temperature is decreasing by 6 degrees, is it going up or down the number line?
- How can you take 6 positive integers away from -4 ? Will you need to add zero pairs to the model? Why or why not?
- What does that look like with integer chips?

6.7.2 Guided Instruction: Subtracting Integers Using a Number Line (continued)

2. Complete the statements to explain how to represent subtraction on a horizontal number line.

- a. To find the difference of $-9 - (-30)$, start by plotting

- ☐ 9
- ☒ -9
- ☐ 30
- ☐ -30

on the number line first.

- b. The subtraction sign indicates moving to the

- ☐ right
- ☒ left

on the number line, but it is followed by

a negative sign with 30, which requires moving in the opposite direction.

- c. Therefore, the answer will be found by moving

- ☐ 9
- ☒ 30

units to the

- ☒ right
- ☐ left

of -9 on the number

line.



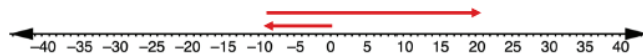
Turn and Talk

Have students consider why the arrow for -30 will move to the right and discuss with a partner. Have some students share their thoughts with the whole class. Consider also asking students to compare the integer chip and number line representations.



What happens when you subtract a negative number? How does this relate to the integer-chip model of representing subtraction?

3. Show the subtraction on the number line below.



The difference of $-9 - (-30)$ is 21.

6.7.3 Collaboration: Relating Integer Addition and Subtraction

Component Overview: With a partner, students choose a context to represent with a number-line model, a subtraction expression, and a related addition equation. Students practice rewriting subtraction equations with the additive inverse of the second addend and represent this on number lines. Partners form groups of four and check each other's work. Finally, students practice representing additive inverses with integer chips.



6.7.3 Collaboration: Relating Integer Addition and Subtraction

Three contexts are given below. Choose one context to work on with your partner. Circle your choice.

Context A	Tamara didn't realize she had a balance of \$19 in her bank account when she used her bank card to purchase a \$50 purse. When the purchase is processed by the bank, what will be her new account balance?
Context B	The elevation of the top floor of a parking garage in Death Valley, California, is 282 feet (ft) below sea level. A car that leaves the parking garage from the top floor loses 72 ft in elevation before reaching the street. What is the elevation of the street?
Context C	Ginny mows lawns for her summer job. Before she started mowing, she spent \$32 on gas for the lawn mower. Then, she spent another \$17 on other supplies she bought at the hardware store. What were Ginny's total expenses?

For the context selected, represent the mathematics in the context in each of the following ways.

Answers will vary. Example is based on Context A.

1. Draw a number line to represent the given context.



2. Write two equations that can be used to find the solution to the question in the context. One equation should use subtraction and the other should use addition.

$$19 - 50 = -31$$

$$19 + (-50) = -31$$



Think-Pair-Share

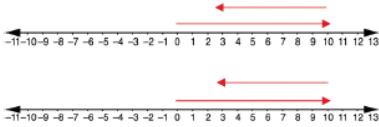
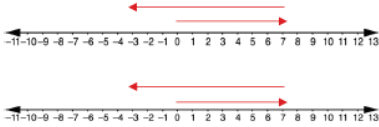
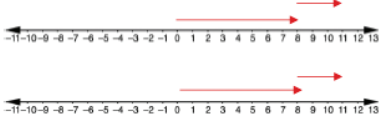
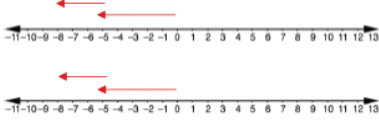
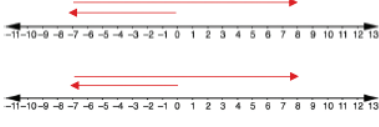
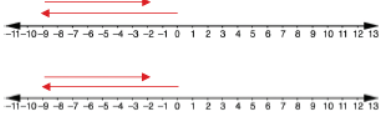
Have students initially complete the problems. Then have students discuss their work with a partner. The discussions will help students gain confidence and understanding.



- How do you represent elevations below sea level on a number line?
- If there's a starting balance of \$19 and Tamara spends more than that, what will her account balance be? How do you represent this on a number line?
- How can you use either subtraction or addition to represent the same scenario?

6.7.3 Collaboration: Relating Integer Addition and Subtraction (continued)

3. Create a group of four by joining with another pair of students. Verify your work by exchanging with the other pair of students.
4. Continue working with the group of four to complete the table.

Example Subtraction Problem	Example Rewritten as an Addition Problem	Number Lines: Show the subtraction on the top number line and the addition on the bottom number line.
$10 - 7 = 3$	$10 + (-7) = 3$	
$7 - 10 = -3$	$7 + (-10) = -3$	
$8 - -3 = 11$	$8 + 3 = 11$	
$-5 - 3 = -8$	$-5 + (-3) = -8$	
$-7 - -15 = 8$	$-7 + 15 = 8$	
$-9 - -7 = -2$	$-9 + 7 = -2$	



What do you notice about the models for each subtraction problem and the expression rewritten with addition?

6.7.3 Collaboration: Relating Integer Addition and Subtraction (continued)

When subtracting integers, problems can be rewritten as an addition problem using the additive inverse of the second integer.

5. Describe one advantage of changing a subtraction problem to an addition problem.

It is easy to forget to go the opposite direction when subtracting. It's easier to add or subtract the absolute values of the addends.

6. Complete the table by rewriting the second integer in the subtraction model as its additive inverse and indicating the sign of the value of either expression. The first row has been completed. Once your group is done, split back into pairs and check your table with a different pair of students.

Visual Model of Subtraction	Visual Model of Addition	Sign of the Answer
		+
		-
		+
		-
		+
		-



Answers will vary for question 5. Consider monitoring for and eliciting specific or targeted responses.

For example, students may share, "When the addition sign is used, I can more easily compare the 'size' or absolute value of each integer to determine the sign."



Students are provided with the first visual model for addition in their workbook. Highlighted in light red, students complete the rest of the second column by rewriting the second integer from the subtraction model as its additive inverse.

6.7.4 Your Turn!

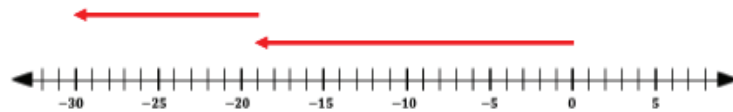
Component Overview: Students are provided with additional practice independently subtracting integers using number lines.



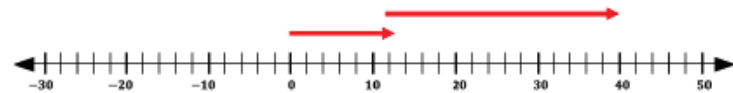
6.7.4 Your Turn!

1. Represent each subtraction statement on a number line and find the value of the expression.

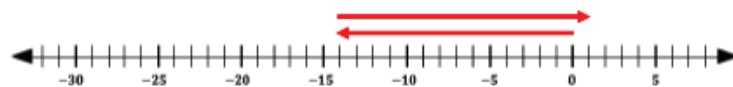
a. $-19 - 11 = -30$



b. $12 - (-28) = 40$



c. $(-14) - (-15) = 1$



Have students rewrite the subtraction expressions as an addition problem. Ask them which expression they would prefer to solve and why.

6.7.5 Wrap-Up: Reflection

Component Overview: Students complete a self-assessment of their understanding of using number lines to subtract integers. Use student reflections to determine which students may need support before the next lesson.



6.7.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can subtract integers using a number line.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
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← Answers will vary. →

2. I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
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← Answers will vary. →



For additional information or to provide alternative means of reflection, prompt students to select one of the following:

- Provide an example based on your location on the number line.
- Provide a mathematical sentence that describes your journey through the lesson. For example, $3 - (-2)$ could represent initially understanding, being unsure, but ending up positive.
- Write question numbers or activities next to where you marked to indicate specific moments in the lesson.